Supplemental Material ***"Resting state EEG and MEG as biomarkers of chronic pain – a systematic review"***

Fig. S1 Sample Sizes of included studies

Fig. S2 Studies over years

Table S1 Risk of Bias assessment of single studies

S1 Search Strategies for Databases and field tags

Fig. S1 Sample Sizes of included studies

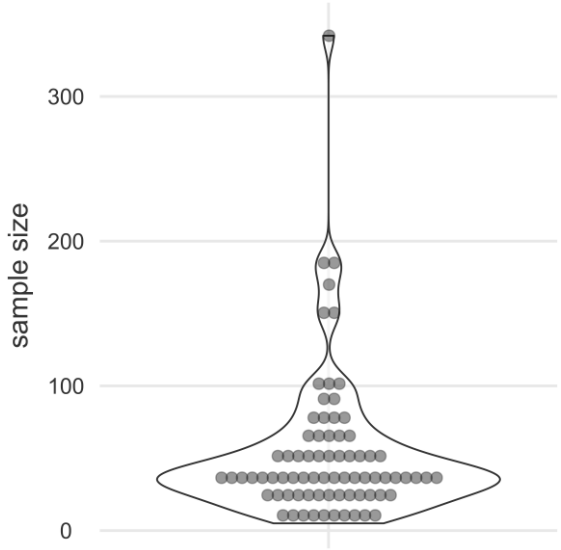


Fig. S2 Studies over years

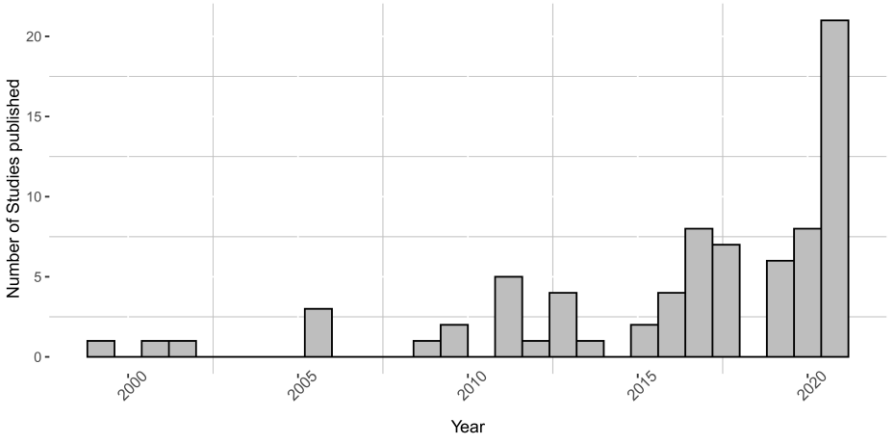


Table S1 Risk of Bias assessment of single studies

	selection of study paricipants				comparability / confounders		outcome data		
	case definition	case representativeness	matching of controls	definition of controls	controlling for anxiety/depression	controlling for any other factor	assessment of outcome (eeg/meg) - blinding	data acquisition / processing (eeg/meg)	appropriate statistical test
Ahn et al 2018	LOW	HIGH	n/a	n/a	HIGH	HIGH	HIGH	LOW	LOW
Barbosa-Torres 2021	LOW	HIGH	n/a	n/a	HIGH	HIGH	HIGH	LOW	LOW
Baroni 2020	LOW	HIGH	LOW	HIGH	HIGH	HIGH	HIGH	LOW	LOW
Bell 2001	LOW	HIGH	HIGH	HIGH	HIGH	LOW	HIGH	LOW	LOW
Bernardi 2020	LOW	HIGH	HIGH	HIGH	HIGH	HIGH	HIGH	LOW	LOW
Camfferman 2017	LOW	HIGH	n/a	n/a	HIGH	HIGH	HIGH	LOW	LOW
Choe 2018	LOW	HIGH	LOW	HIGH	HIGH	HIGH	HIGH	LOW	LOW
Day 2020	LOW	HIGH	n/a	n/a	HIGH	HIGH	LOW	LOW	LOW
Day 2021	LOW	HIGH	n/a	n/a	HIGH	HIGH	HIGH	LOW	LOW
deMelo 2020	LOW	HIGH	n/a	n/a	LOW	LOW	HIGH	LOW	LOW
deMelo 2021	LOW	HIGH	n/a	n/a	HIGH	HIGH	HIGH	LOW	LOW
DeRidder 2017	LOW	HIGH	LOW	HIGH	HIGH	HIGH	HIGH	LOW	LOW
deVries 2013	LOW	LOW	LOW	HIGH	HIGH	HIGH	HIGH	LOW	LOW
DiPietro 2018	LOW	HIGH	LOW	HIGH	HIGH	HIGH	HIGH	HIGH	LOW
Fallon 2018	LOW	HIGH	LOW	HIGH	HIGH	HIGH	HIGH	LOW	LOW
Fauchon 2021	LOW	HIGH	LOW	LOW	HIGH	LOW	HIGH	LOW	LOW
Feng 2021	LOW	HIGH	n/a	n/a	HIGH	HIGH	HIGH	LOW	LOW
Ferdek 2019	LOW	HIGH	LOW	HIGH	HIGH	HIGH	HIGH	LOW	LOW
Freye 2006	LOW	HIGH	n/a	n/a	HIGH	HIGH	HIGH	HIGH	LOW
Gonzalez-Roldan 2016	LOW	HIGH	LOW	HIGH	LOW	LOW	HIGH	LOW	LOW
Gonzalez-Villar 2020	LOW	HIGH	LOW	LOW	HIGH	HIGH	HIGH	LOW	LOW
Gram 2017	LOW	LOW	n/a	n/a	HIGH	HIGH	HIGH	LOW	LOW
Graversen 2012	LOW	HIGH	LOW	LOW	HIGH	HIGH	HIGH	LOW	HIGH
Hargrove 2010	LOW	HIGH	LOW	HIGH	HIGH	HIGH	HIGH	LOW	HIGH
Heitmann 2022	LOW	HIGH	n/a	n/a	HIGH	LOW	HIGH	LOW	LOW
Hsiao 2017	LOW	LOW	LOW	LOW	LOW	LOW	HIGH	LOW	LOW
Hunter 2009	LOW	HIGH	n/a	n/a	LOW	HIGH	HIGH	LOW	LOW
Iwatsuki 2021	LOW	HIGH	HIGH	LOW	HIGH	HIGH	HIGH	LOW	HIGH
Jensen 2013 *[45]**	LOW	HIGH	HIGH	HIGH	HIGH	HIGH	HIGH	LOW	LOW
Jensen 2013 *[42]**	LOW	HIGH	n/a	n/a	HIGH	HIGH	HIGH	LOW	LOW
Jensen 2021	LOW	LOW	n/a	n/a	HIGH	LOW	HIGH	LOW	LOW
Juel 2017	LOW	HIGH	n/a	n/a	HIGH	HIGH	HIGH	LOW	HIGH
Kayiran 2010	LOW	LOW	n/a	n/a	HIGH	HIGH	HIGH	LOW	HIGH
Kim 2019	HIGH	HIGH	LOW	LOW	HIGH	HIGH	HIGH	LOW	LOW

Kim 2020	HIGH	HIGH	HIGH	LOW	HIGH	HIGH	HIGH	LOW	LOW
Kim 2021	LOW	LOW	n/a	n/a	HIGH	HIGH	HIGH	LOW	HIGH
Kisler 2020	LOW	HIGH	LOW	LOW	HIGH	HIGH	HIGH	LOW	LOW
Lee 2018	LOW	HIGH	n/a	n/a	HIGH	HIGH	HIGH	LOW	LOW
Levitt 2020	LOW	HIGH	LOW	LOW	HIGH	HIGH	HIGH	LOW	LOW
Lim 2016	LOW	HIGH	LOW	HIGH	HIGH	HIGH	HIGH	LOW	HIGH
Lopez 2015	LOW	HIGH	LOW	HIGH	HIGH	HIGH	HIGH	LOW	HIGH
Martin-Brufau 2021	LOW	LOW	LOW	HIGH	HIGH	HIGH	HIGH	LOW	LOW
May 2021	LOW	HIGH	LOW	LOW	HIGH	HIGH	HIGH	LOW	LOW
Meneses 2016	LOW	HIGH	HIGH	LOW	LOW	HIGH	HIGH	LOW	LOW
Michels 2011	LOW	HIGH	HIGH	LOW	LOW	LOW	HIGH	LOW	LOW
Nomura 1999	LOW	HIGH	LOW	HIGH	HIGH	HIGH	HIGH	LOW	HIGH
Oga 2002	HIGH	HIGH	n/a	n/a	HIGH	HIGH	HIGH	LOW	LOW
Olesen 2011	LOW	HIGH	LOW	LOW	HIGH	HIGH	HIGH	LOW	LOW
Parker 2021	LOW	HIGH	n/a	n/a	HIGH	HIGH	HIGH	LOW	LOW
Patel 2020	HIGH	HIGH	HIGH	HIGH	HIGH	HIGH	HIGH	LOW	LOW
Prichep 2011	LOW	HIGH	n/a	n/a	HIGH	HIGH	HIGH	LOW	HIGH
Prichep 2018	LOW	HIGH	LOW	LOW	HIGH	HIGH	HIGH	LOW	LOW
Prinsloo 2017	LOW	HIGH	LOW	LOW	HIGH	LOW	HIGH	LOW	LOW
Raghu 2021	LOW	HIGH	LOW	LOW	HIGH	HIGH	LOW	LOW	LOW
Sarnthein 2006	LOW	HIGH	LOW	LOW	HIGH	HIGH	HIGH	LOW	LOW
Schmidt 2012	LOW	HIGH	LOW	LOW	HIGH	HIGH	HIGH	LOW	LOW
Schmidt 2015	LOW	HIGH	n/a	n/a	HIGH	HIGH	HIGH	LOW	LOW
Stefanie 2011	LOW	HIGH	LOW	HIGH	LOW	HIGH	HIGH	LOW	LOW
Stern 2006	LOW	LOW	LOW	LOW	HIGH	HIGH	HIGH	LOW	HIGH
Sufianov 2014	LOW	HIGH	HIGH	HIGH	HIGH	HIGH	HIGH	HIGH	LOW
TaDinh 2019	LOW	HIGH	LOW	LOW	HIGH	HIGH	HIGH	LOW	LOW
Teixeira 2021 *[105]**	LOW	HIGH	LOW	LOW	HIGH	HIGH	HIGH	LOW	LOW
Teixeira 2021 *[106]**	LOW	HIGH	n/a	n/a	LOW	LOW	HIGH	LOW	HIGH
Thibaut 2017	LOW	HIGH	HIGH	HIGH	HIGH	HIGH	HIGH	LOW	LOW
Uygur-Kucukseymen 2020	LOW	HIGH	n/a	n/a	HIGH	LOW	HIGH	LOW	LOW
vandenBroeke 2013	LOW	HIGH	HIGH	HIGH	HIGH	HIGH	HIGH	LOW	LOW
Vanneste 2017	LOW	LOW	LOW	HIGH	HIGH	HIGH	HIGH	LOW	LOW
Vanneste 2018	LOW	HIGH	HIGH	HIGH	HIGH	HIGH	HIGH	LOW	LOW
Vanneste 2021	LOW	HIGH	HIGH	LOW	HIGH	HIGH	HIGH	LOW	LOW
Villafaina 2019	LOW	HIGH	LOW	LOW	HIGH	HIGH	HIGH	LOW	LOW
Witjes 2021	LOW	HIGH	HIGH	LOW	HIGH	HIGH	HIGH	LOW	LOW
Xavier 2021	LOW	HIGH	LOW	LOW	HIGH	HIGH	HIGH	LOW	LOW
Yüksel 2019	LOW	HIGH	HIGH	HIGH	HIGH	HIGH	HIGH	LOW	HIGH
Zhou 2018	LOW	HIGH	LOW	HIGH	LOW	HIGH	HIGH	LOW	LOW
Zolezzi 2021	HIGH	HIGH	n/a	n/a	HIGH	HIGH	HIGH	LOW	LOW
Zortea 2021	LOW	HIGH	LOW	LOW	HIGH	LOW	HIGH	LOW	LOW

** numbers in brackets refer to references in main text*

S1 Search Strategies for Databases and field tags

MEDLINE (via PubMed): “all fields”

((fibromyalgia) OR (CRPS OR complex regional pain syndrome) OR (temporomandibular disorder) OR (burning mouth syndrome) OR (irritable bowel syndrome) OR (trigeminal neuralgia) OR (chronic painful polyneuropathy) OR (postherpetic neuralgia) OR (chronic painful radiculopathy) OR (chronic AND pain)) AND (EEG OR electroencephalogra OR MEG OR magnetoencephalogra*)*

Web of Science Core Collection (via Web of Science): “all fields”

((fibromyalgia) OR (CRPS OR complex regional pain syndrome) OR (temporomandibular disorder) OR (burning mouth syndrome) OR (irritable bowel syndrome) OR (trigeminal neuralgia) OR (chronic painful polyneuropathy) OR (postherpetic neuralgia) OR (chronic painful radiculopathy) OR (chronic AND pain)) AND (EEG OR electroencephalogra OR MEG OR magnetoencephalogra*)*

Cochrane Central Register of Controlled Trials (via Ovid): “all fields”

((fibromyalgia) OR (CRPS OR complex regional pain syndrome) OR (temporomandibular disorder) OR (burning mouth syndrome) OR (irritable bowel syndrome) OR (trigeminal neuralgia) OR (chronic painful polyneuropathy) OR (postherpetic neuralgia) OR (chronic painful radiculopathy) OR (chronic AND pain)) AND (EEG OR electroencephalogra OR MEG OR magnetoencephalogra*)*

EMBASE (via Ovid): “all fields”, results limited to “articles” and “articles in press”

((fibromyalgia) OR (CRPS OR complex regional pain syndrome) OR (temporomandibular disorder) OR (burning mouth syndrome) OR (irritable bowel syndrome) OR (trigeminal neuralgia) OR (chronic painful polyneuropathy) OR (postherpetic neuralgia) OR (chronic painful radiculopathy) OR (chronic AND pain)) AND (EEG OR electroencephalogra OR MEG OR magnetoencephalogra*)*